

Variance and Covariance

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1 Setup

Throughout this lesson, fix a probability space $(\Omega, \mathcal{F}, P)$ and let $L^2(P) = \{X : \Omega \rightarrow \mathbb{R} \mid \mathbb{E}[X^2] < \infty\}$ denote the Hilbert space of square-integrable random variables. All random variables below are assumed to lie in $L^2(P)$, which guarantees the existence of the moments $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and (by Cauchy–Schwarz) $\mathbb{E}[XY]$ for $X, Y \in L^2(P)$.

2 Definitions

Definition 1 (Variance). The *variance* of $X \in L^2(P)$ is

$$\text{Var}(X) := \mathbb{E}[(X - \mathbb{E}[X])^2].$$

By expanding the square and applying linearity of expectation,

$$\text{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2.$$

The *standard deviation* is $\sigma_X = \sqrt{\text{Var}(X)}$.

Definition 2 (Covariance). The *covariance* of $X, Y \in L^2(P)$ is

$$\text{Cov}(X, Y) := \mathbb{E}[(X - \mathbb{E}[X])(Y - \mathbb{E}[Y])] = \mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y].$$

Note $\text{Cov}(X, X) = \text{Var}(X)$.

Definition 3 (Correlation). If $\sigma_X, \sigma_Y > 0$, the (*Pearson*) *correlation coefficient* of X and Y is

$$\rho_{X,Y} := \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}.$$

By Cauchy–Schwarz applied to the centered variables, $\rho_{X,Y} \in [-1, 1]$.

3 Elementary properties

Lemma 1 (Bilinearity and symmetry of covariance). For $a, b, c \in \mathbb{R}$ and $X, Y, Z \in L^2(P)$:

1. $\text{Cov}(X, Y) = \text{Cov}(Y, X)$.
2. $\text{Cov}(aX + b, Y) = a \text{Cov}(X, Y)$.
3. $\text{Cov}(X + Z, Y) = \text{Cov}(X, Y) + \text{Cov}(Z, Y)$.

Corollary 2 (Affine scaling of variance). For $a, b \in \mathbb{R}$, $\text{Var}(aX + b) = a^2 \text{Var}(X)$.

4 Variance of a sum (main theorem)

Theorem 3 (Variance of a sum). For $X, Y \in L^2(P)$,

$$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2 \text{Cov}(X, Y).$$

Proof. Let $\mu_X = \mathbb{E}[X]$ and $\mu_Y = \mathbb{E}[Y]$. By linearity of expectation, $\mathbb{E}[X + Y] = \mu_X + \mu_Y$. Applying the definition of variance to $X + Y$,

$$\begin{aligned} \text{Var}(X + Y) &= \mathbb{E} \left[((X + Y) - \mathbb{E}[X + Y])^2 \right] \\ &= \mathbb{E} \left[((X - \mu_X) + (Y - \mu_Y))^2 \right] \\ &= \mathbb{E} \left[(X - \mu_X)^2 + 2(X - \mu_X)(Y - \mu_Y) + (Y - \mu_Y)^2 \right]. \end{aligned}$$

By linearity of expectation,

$$\begin{aligned} \text{Var}(X + Y) &= \mathbb{E}[(X - \mu_X)^2] + 2\mathbb{E}[(X - \mu_X)(Y - \mu_Y)] + \mathbb{E}[(Y - \mu_Y)^2] \\ &= \text{Var}(X) + 2 \text{Cov}(X, Y) + \text{Var}(Y), \end{aligned}$$

where the last equality uses the definitions of variance and covariance. □

Corollary 4 (n -term version). For $X_1, \dots, X_n \in L^2(P)$,

$$\text{Var} \left(\sum_{i=1}^n X_i \right) = \sum_{i=1}^n \text{Var}(X_i) + 2 \sum_{1 \leq i < j \leq n} \text{Cov}(X_i, X_j).$$

In particular, if the X_i are pairwise uncorrelated, then $\text{Var}(\sum X_i) = \sum \text{Var}(X_i)$.

Remark. Independence of X and Y implies $\mathbb{E}[XY] = \mathbb{E}[X]\mathbb{E}[Y]$, hence $\text{Cov}(X, Y) = 0$. The converse is false: take $X \sim \text{Uniform}(\{-1, 0, 1\})$ and $Y = X^2$; then $\text{Cov}(X, Y) = 0$ but Y is a deterministic function of X .

5 Worked example

Bernoulli. Let $X \sim \text{Bernoulli}(p)$. Then $\mathbb{E}[X] = p$, $\mathbb{E}[X^2] = p$, so

$$\text{Var}(X) = p - p^2 = p(1 - p).$$

The variance is maximized at $p = 1/2$.